

Data-Driven Strategies for Detecting and Sampling Misrepresented Subgroups

Giacomo Lancia¹, Fulvia Mecatti², Eva Riccomagno^{1*}

^{1*}Department of Mathematics, Università di Genova, Via Dodecaneso, 35, 16146, Genova, Italy.

²Department of Sociology and Social Research, Università degli studi di Milano-Bicocca, Via Bicocca degli Arcimboldi, 8, 20126, Milano, Italy.

*Corresponding author(s). E-mail(s): eva.riccomagno@unige.it;

Abstract

Economic policy research often investigates population well-being, with particular attention to the links between unequal living conditions, low education, and social exclusion. Sample surveys such as European Union Statistics on Income and Living Conditions (EU-SILC) are widely used for this purpose and inform public policy, yet their sampling designs may fail to adequately represent rare, hard-to-sample, or under-covered subgroups.

We propose a generalisable data-driven integrated workflow to detect outliers in survey data that may signal under-represented subgroups. Identified groups can then be characterised to inform targeted resampling strategies that improve survey inclusiveness. An empirical application using the 2019 EU-SILC data for the Italian region of Liguria shows that citizenship, material deprivation, large household size, and economic vulnerability are indicators of under-representation.

Keywords: Socio-economic vulnerability, surveyed data, under-represented subgroups, unsupervised learning

1 Introduction

As our world becomes increasingly interconnected, the role of poverty and inequalities in wealthy societies becomes ever more important. The European Union Statistics on Income and Living Conditions programme ([European Union Statistics on Income and Living Conditions \(EU-SILC\)](#)) Eurostat [9] is the reference source for comparative

statistics on income distribution and social inclusion in the European Union and is used for policy monitoring within the Open method of coordination. Representation equity and the production of equitable statistics are key objectives of official statistics and necessary conditions for data-driven and unbiased policy-making. They involve fostering the visibility of specific population subgroups in the data, which may hide underserved communities. EU-SILC, being a survey intended for general users and purposes, may pose challenges in both respects. This paper is concerned with two main objectives:

1. To apply advanced data-driven methods to EU-SILC cross-sectional data in order to detect hidden, rare, or otherwise under-represented population subgroups — such as vulnerable or marginalised communities — that may require targeted policy attention.
2. To discuss state-of-the-art sampling strategies for the effective integration of EU-SILC data, specifically targeting misrepresented subgroups and population segments not sufficiently captured in the EU-SILC sample.

Objective 1 goes beyond the inferential goals of sample representativeness and unbiased estimation, which are ensured by relying on official statistics such as EU-SILC data. Instead, it concerns a potential scarcity of data for particular layers of the population under study. Indeed, good performances for overall estimands do not ensure adequate representation of rare and vulnerable population segments, as well as unknown or unplanned subgroups that are relevant but under-represented in the sample data. In this paper, we shall use the general term “*misrepresented subgroups*” to denote these cases.

Both objectives are related to critical concerns such as hard-to-sample populations and undercoverage of the sampling frame, which are well known in survey sampling methodology and practice and have produced a consistent and well-established body of literature. Both objectives can be framed as a search for informative observations, such as high leverage points and outliers, for example by adopting an information based approach (van Cranenburgh and Bliemer [25]) and adapting optimal design strategies, such as D -optimal design of experiments as in Deldossi et al. [7] and Wang et al. [27].

However, although such literatures can help address objective 2, a literature gap emerges with respect to objective 1. The purpose of detecting misrepresented population subgroups may, in fact, extend beyond known and pre-defined subgroups, for instance, population strata not included as eligible for the survey, or unknown and hard-to-predict subgroups. While recent literature focuses mainly on health studies (e.g. Törnävä et al. [24]), this paper fills the methodological gap in social research by targeting misrepresented subgroups in EU-SILC data, with methods extendable to other contexts and datasets

This paper is motivated by a real case study proposed by the local government of the Italian Liguria region (see sub-section 2.1). As a consequence, the objectives above also entail an improved granularity in order to reflect geographical heterogeneity. A high-quality large-scale probability survey of households, as EU-SILC, ensures accurate estimates at the global EU level and for each nation as a whole. However, it can be inadequate to meet the data demands at the sub-national level and for key population

subgroups, which can result in data inequities with subsequent lack of information for unbiased policy making.

The paper is organised as follows: Section 2 provides a description of the Ligurian subsample of the 2019 EU-SILC microdata. It addresses the issue of misrepresented subgroups in this data and the reasons why this lack of representation was expected, as well as the sampling design context. Section 3 describes the methodology employed, while Section 4 presents the results obtained. To mitigate the problem of missing misrepresented subgroups, Section 5 presents a simulation study to evaluate integrative sampling schemes to append to the existing survey. Section 6 concludes the paper with a discussion. The supplementary material includes an extended background on the context and detailed information on the used EU-SILC in Appendix A; additional technical details intended for readers without a background in supervised learning in Appendix B while or a background on the multi-frame approach we refer to Mecatti [14]; details on the empirical analysis on Ligurian 2019 EU-SILC in Appendix C; details of the simulation settings used in Section 5 and information on code availability in Appendix D.

2 Background, Context and Data

EU-SILC data are used by European, national and local administrations for various purposes, such as comparing socio-economic conditions across European countries or informing local or national policies (see Appendix A). When using EU-SILC microdata to monitor living conditions in Europe, critically assessing data quality is essential. Recent studies have documented significant cross-national variation, especially in income variables (Trindade and Goedemé [22]), with quality issues flagged by Hauser [10]. A 2020 analysis by Commission and Eurostat [5] revealed substantial non-compliance with Eurostat’s guidelines in several member states, hampering comparability. These issues are particularly concerning when data inform local policies, underscoring the need for more robust approaches to representation equity. The EU-SILC sampling scheme remained unchanged until 2023, when a 2024 questionnaire upgrade aimed to improve data reliability (Wirth and Pforr [26]).

2.1 The Liguria Project

This study was initiated in response to an internal need of the Liguria Region to evaluate the impact of newly implemented policies, primarily aimed at promoting a *Pact for Residential Stability* and *Family Support* measures; for details refer to Appendix A. In assessing the effectiveness of these measures, the Region faced a lack of specific internal data and found the EU-SILC 2019 Liguria dataset insufficient for this purpose, as those households were not adequately represented in the sample. Consequently, it prompted this study to examine in greater detail the case of the misrepresented population subgroups within the EU-SILC cross-sectional sample specific for Liguria.

2.2 2019 EU-SILC: Sample Design and Data

The general guidelines for the 2019 EU-SILC survey rigorously define the reference population and eligibility criteria as

All private households and their current residing members at the time of data collection, excluding collective households and institutions. All private households and all persons aged 16 and over within the household are eligible for the survey operation

Children under the age of 16 are covered indirectly through household-level information reported by adult members. At the same time, the guidelines grant the National Statistical Offices of all participating Member States in the EU-SILC programme, who are responsible for data collection, considerable flexibility in their choice of sampling design. This flexibility is exercised with a focus on delivering timely and internationally comparable cross-sectional data, while implementing appropriate procedures, in line with each office’s own best practices, to maximise response rates.

A nationally representative probability sample of the target population is required, both at the household and at individual person level. The sampling frame and sample selection ensure a known and positive (first-order) inclusion probability. The national minimum effective sample size is stated based on practical considerations and precision requirements for the released estimates, e.g. for Italy 2019, a cross-sectional sample of 7250 households and of 15500 persons aged 16 and over to be interviewed.

The EU-SILC database offers a rich array of information of various kinds. Our case study focuses on a subset of variables selected from the questionnaire concerning the living conditions and social exclusion of Liguria’s population. Specifically, to prioritise the identification of macroscopic trends, only variables uniformly applicable across the population were considered. Variables that could introduce structurally missing data, due to being limited to specific respondent categories, such as pensioners, renters, households with minors, or recipients of employment-related benefits, were excluded. The resulting dataset is complete, requiring no imputation or adjustment for missing data. This led to the selection of 19 discrete variables (ordinal, categorical, and dichotomous), while quantitative and continuous variables self-reported and required discretisation were omitted. A full description of the selected variables is provided in Appendix A.

The core contribution of this work, presented in Sections 3 and 4, addresses objective 1 of Section 1, through the identification of outliers within the empirical distribution functions of the selected study variables, which are interpreted as red flags for potentially misrepresentation of population subgroups warranting further inspection. This intuition was prompted by a comparison of the representativeness of EU-SILC data with respect to a complete administrative source of socio-demographic data concerning tax records.

2.3 Liguria Complete Tax Administrative Data

2.3.1 2019 Tax Declaration Dataset

From the Italian national tax records warehouse, we extracted the portion of specific administrative information concerning the Ligurian region in 2019, hereafter referred to as the *Tax Declaration Dataset*. This dataset is non-public, subject to data protection regulations, and includes anonymised tax records. It offers a comprehensive overview of the income and economic status of most households in the region. It contains a total of 1.216.747 individual tax declarations, corresponding to 731.537 distinct

households. When compared to the official population census data provided by ISTAT, this coverage accounts for approximately 96.5% of all Ligurian households, making it a highly representative and detailed data source. Because of its quasi-complete coverage, this dataset provides reliable demographic and economic indicators, such as sex, age, employment status, annual income, number of dependent children, and number of dependent disabled children.

2.3.2 Comparative Analysis of EU-SILC and Tax Declaration Datasets

We leverage the Tax Declaration Dataset to identify potential mismatches and misrepresented demographic subgroups in EU-SILC, that may be misrepresented or missed altogether in the survey. We focus on two variables available in both sources: women aged 15–55 and children aged 0–17 per household. Figure 1 presents comparative histograms accounting for adjusted design weights (Section 3.5).

Notable discrepancies emerge. For children aged 0–17, EU-SILC fails to capture households with more than five children (0.1% prevalence), while slightly over-representing those with one to four children. Similarly, households with more than five women aged 15–55 are underrepresented (a highly rare group accounting for at most 0.001% of the Ligurian population), and households with no women aged 15–55 account for only 50% of the EU-SILC sample versus 72% in the tax data. EU-SILC tends to slightly over-represent households containing between two and five women aged 15–55. Furthermore, some subgroups well captured in administrative data are entirely absent from EU-SILC — for instance, dependent children with disabilities.

These findings support framing the identification of misrepresented subgroups as an outlier detection problem, offering a systematic approach to uncovering groups underserved by EU-SILC data.

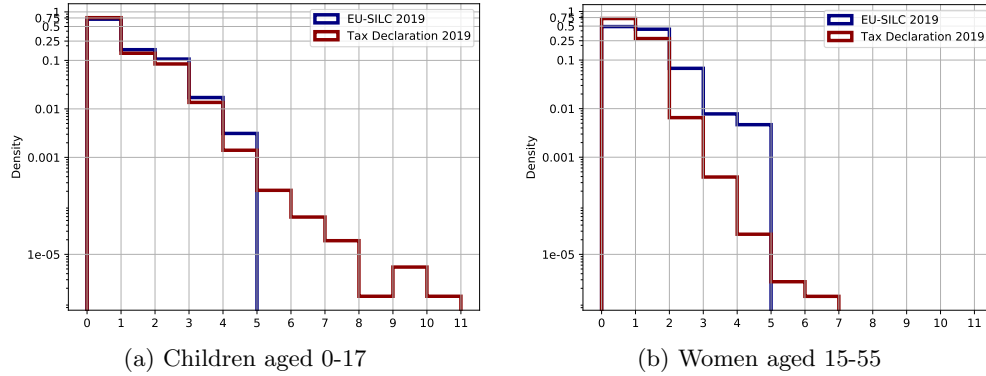


Fig. 1: Comparison of household-level distributions for children aged 0-17 and women aged 15-55, in the Ligurian subsample of the 2019 EU-SILC survey and the administrative Tax Declaration Dataset for Liguria (2019).

3 Detecting misrepresented subgroups

We adapt three outlier detection techniques to identify potentially misrepresented subgroups in the Ligurian population. We work with unlabelled data, and construct a binary labelling separating typical observations —defined as the most recurrent EU-SILC patterns— from atypical ones. This approach exploits outlier detection as an exploratory, unsupervised method to identify subpopulations layers which are potentially scarcely represented. While outlier profiles may not necessarily coincide with all misrepresented population subgroups, they are highlighted as indicators of potential information gaps.

The absence of ground truth limits evaluation to internal validation, as no fully consistent external reference dataset is available both in terms of coverage and temporal alignment. Also the tax data in Section 2.3.2 only partially overlaps with this analysis. We therefore use cross-validation and compare multiple models for consistency. Observations flagged as outliers across multiple folds and model runs should be interpreted as systematic signals of divergence from the majority.

3.1 Anomaly detection models

We implement both univariate and multivariate models. Those presented below yielded satisfactory results, although alternative approaches could also be considered.

As a unidimensional strategy, we propose an *entropy-based score*, which exploits the property of entropy as an aggregate measure of information, typically suggesting how infrequent an observation is. The rationale for including this strategy lies in its ability to analyse each variable independently and as unassociated. As a result, the identification of misrepresented subpopulations consists of aggregating those variables that both predominantly influence the anomaly detection process and stand out as markedly atypical.

For a multivariate perspective, we considered both standard statistical learning and ML-based approaches, focusing on two reconstruction-based models: [Kernel Principal Component Analysis \(KPCA\)](#) and [Auto-Encoder \(AE\)](#). Through an inverse transformation, either exact or approximate, both approaches encode high-dimensional data into a lower-dimensional latent space and reconstruct the original input, optimising to preserve as much variance as possible.

In anomaly detection, the reconstruction error serves as the outlier signal: observations with high reconstruction error are flagged as atypical, as their patterns deviate from those learned from the majority of the data.

3.1.1 Entropy Score

Our approach consists of two steps: first, compute an entropy-based score for each variable to identify those most informative ones and likely to containing outliers; second, use information content to pinpoint the specific categories best representing the target anomalies.

For a categorical variable X with k categories and relative frequencies p_i for $i = 1, \dots, k$, we considered the following score based on the Shannon Entropy Index

$$\Gamma_1 = 1 + \frac{1}{\log k} \sum_{i=1}^k p_i \log p_i \in [0, 1], \quad (1)$$

with higher values indicating a greater likelihood of misrepresented categories. We computed Γ_1 across the 19 variables in Appendix A and select the five with highest score (see Section 3.2). We then assess the information content of each category of these five variables using $-\log p_i$. This helps pinpoint potentially misrepresented groups within those variables. Further technical details are in Appendix B.1

3.1.2 Kernel Principal Component Analysis

A non-linear extension of [Principal Component Analysis \(PCA\)](#) is [KPCA](#) which generalises PCA by implicitly mapping the original data into a high-dimensional feature space, called the Φ -space, through a nonlinear transformation Φ learned from the data. A linear [PCA](#) is then performed in the Φ -space (e.g. Hastie et al. [11]).

Observations with poor reconstruction quality are likely to be outliers, as they do not conform well to the learned data patterns. The *reconstruction error* is evaluated in the feature space as

$$\text{RE}_{\text{KPCA}}(\Phi(\mathbf{z})) = \|\Phi(\mathbf{z}) - \hat{\Phi}(\mathbf{z})\|^2$$

where $\mathbf{z}$ is the original data point and $\hat{\Phi}(\mathbf{z})$ denotes the projection of $\Phi(\mathbf{z})$ onto the subspace capturing 95% of the variance in the transformed feature space.

The transformation Φ is typically determined via kernel matrices. We use a Hamming metrics-based kernel matrix, ensuring that the constructed kernel is positive semidefinite. The eigen decomposition of the kernel matrix enables the determination of the transformations required for decomposing and reconstructing the data. Further technical details are provided in Appendix B.2.

3.1.3 Auto-Encoder

An Auto-Encoder ([AE](#)) is a neural network designed to learn efficient representations of input data in an unsupervised manner (see Aggarwal [1]). It consists of two components: an *encoder* and a *decoder*, typically designed as two feed-forward neural networks (usually, multi-layer perceptrons) with the scope of compressing and subsequently decompressing data. For the outlier detection problem, [AE](#)-based strategies have been proposed in several applications, more recently by Abhaya and Patra [2], Torabi et al. [23] among others. The underlying assumption is that the [AE](#) learns how to accurately compress and reconstruct *typical* data but will yield a high reconstruction error for anomalous inputs not well represented in the training distribution. Thus, the reconstruction error serves as an unsupervised anomaly score.

In this work, to avoid overfitting and unnecessary complexity, we opted for a shallow architecture: both the encoder and decoder consist of two hidden layers each as

in Aggarwal [1]. The encoder’s first layer is a dense (fully connected) layer with a hyperbolic tangent activation function, which reduces the dimensionality of the input data by half. The second layer, identical in structure, further reduces the dimensionality by half. Consequently, the embedding space has a dimensionality equal to one-quarter of the original input. The decoder mirrors the encoder in structure. The first layer is a dense layer that doubles the size of the embedding vector. The second layer is also dense and uses a softplus activation function, following Glorot et al. [9], which smoothly transforms the outputs and restores the vector to its original dimensionality. In other words, the second layer applies a smoothing nonlinear function, i.e., the softplus function, defined as $\log(1 + \exp(\cdot))$, to ensure all outputs remain positive while keeping the overall structure of the data. Finally, since the input data are discrete, we include an additional rounding step when evaluating the reconstruction error. This post-processing step ensures consistency with the original data format while preserving the benefits of smooth training dynamics.

3.2 Outlier Detection

While both the Entropy-Based Score and the reconstruction error-based methods provide meaningful scores, they do not supply a threshold for classifying observations as outliers. To overcome this issue, we adopted an unsupervised strategy.

For the Entropy-Based Score, we apply k -means clustering with $k = 2$ directly to the score distribution. This yields a data-driven threshold on Γ_1 . For the [KPCA](#) and [AE](#) methods, we proceed analogously, but at the instance level rather than at the variable level. Specifically, we apply k -means clustering with $k = 2$ to the reconstruction error scores, in line with the intuition that items with higher reconstruction errors are more likely to be outliers. In fact, clustering based on a one-dimensional score, whether Γ_1 or a reconstruction-based error, separates the data into two groups, with the cluster having the higher centroid (i.e., larger average score) labelled as outliers.

Although k -means implicitly assumes the presence of cluster structure by maximising between-cluster separation, additional analysis based on Gaussian Mixture Model (GMM) supports this choice. In particular, model comparison indicates that a two-component solution provides a substantially better fit than a single-component model, suggesting that the data are better represented by multiple latent groups rather than by a single homogeneous distribution with extreme values.

Finally, a sensitivity analysis on the threshold selection was conducted, showing that the proposed k -means-based threshold is stable and robust in identifying the underlying groups. Further details are provided in [Section B.5](#).

3.3 Validation process

Given the absence of ground truth labels in our unsupervised setting, model evaluation relied on three procedures: stability validation, internal validation, and cross-validation. *Stability validation* allows the assessment of the robustness of the clustering outcome and evaluates whether small perturbations in the data lead to consistent clustering solutions. We adopted a leave-one-out strategy: at each iteration, one item was excluded and clustering was performed on the remaining items. The obtained

clusters were compared across iterations using the [Matthews Correlation Coefficient \(MCC\)](#) introduced in Matthews [13]. Primarily applied in supervised learning, MCC can also be used to provide a balanced measure of agreement between clustering assignments (see Chicco and Jurman [6]). It measures the consistency between cluster labels derived from cross-validation and those obtained from the full-data model. A high MCC value indicates supports robustness. This procedure allowed us to evaluate the robustness of the 2-cluster K-means algorithm in consistently identifying outlier structures across the dataset.

To complement stability validation, an *internal validation* was performed by applying 10-fold cross-validation to assess the generalisability of the clustering method. The data are partitioned into subsets. At each iteration, one fold was held out while the remaining nine trained the model. The held-out fold then served as a validation set for outlier detection. Since folds are non-overlapping, this yielded clustering assignments for each excluded subset, which were compared against those from the full dataset. A robust method should produce consistent assignments across folds, as removing a single fold should not substantially alter the clustering structure.

For *cross-validation*, a 10-fold procedure was used to evaluate each model’s ability to detect novel outliers. Following standard practice (see Marques et al. [16]), the unsupervised framework is reformulated as a one-class classification task, where each detector assigns a continuous anomaly score reflecting deviation from normality. Within each split, the training fold defines the inlier class and serves to construct a nearest-neighbour-based reference score distribution. We excluded the top 10% highest-scoring observations for robustness. Outlier scores for each validation fold are computed using the model fitted on the corresponding training data and compared against this reference distribution. Classification thresholds are determined by applying a two-cluster k-means algorithm solely to the training scores. Predictive performance is assessed using [Area Under the Receiver Operating Characteristic Curve \(AUROC\)](#).

3.4 Variable Inspection

To extrapolate from the proposed multivariate unsupervised methodology, which variables (and the corresponding population subgroups) contribute most to the anomalous characteristics of the detected outliers, we employed the *Feature Importance* algorithm introduced by Breiman [4]. This method was originally presented to inspect supervised models. It operates by disrupting the relationship between each input variable and the target variable, assessing the extent to which the model’s predictions degrade when the association is broken.

In our application, each variable is shuffled one at a time, while all other variables remain unchanged. The modified data are then fed into the anomaly detection models, and their scores are passed to the fitted clustering model. The impact due to the shuffling of a variable is assessed by measuring the *importance score*, namely, the change in performance, quantified as the difference in the *accuracy score* (i.e., the fraction of correct predictions) for the identification of anomalous items. This difference is used as an estimate of the variable’s importance. The greater the importance assigned to a variable, the more influence it exerts on the outlier detection process. To ensure

stability, each variable was permuted 30 times. The mean importance score and the corresponding 95% confidence interval for each feature were estimated via jackknife resampling. This approach provides a robust quantification of each feature’s influence on the detection of anomalies.

3.4.1 Exploring Under-represented Subgroups

To identify potential under-represented subpopulations among the items previously labelled as outliers, we perform an additional clustering step restricted to this subset. Given that the structure of the data may not reside in a convex space, we adopt spectral clustering, a technique specifically suited for identifying complex, non-convex cluster geometries (see Hastie et al. [11]). It is important to note that this secondary clustering procedure is applied exclusively to the outliers identified by the unsupervised anomaly scores derived from the [KPCA](#) and [AE](#). The spectral clustering is executed solely on the set of outlier items, with the aim of discovering internal structure or subgroupings within this anomalous region.

To interpret the resulting clusters, we focus on visualising their most representative members, specifically, the *medoids* of each cluster. By Struyf et al. [20] the medoid is defined as the observation within a cluster that minimises the average distance to all other points in the same cluster, thereby offering an interpretable exemplar of the cluster’s characteristics. The optimal number of clusters is determined by selecting the configuration that maximises the *Silhouette Score*, introduced by Rousseeuw [18], over a predefined range of possible cluster counts. For consistency with our earlier methodology, we construct the affinity matrix for spectral clustering using the same Hamming distance-based kernel employed in the [KPCA](#). The Hamming distance is also used in the computation of silhouette scores.

3.5 Incorporating Sample Design Weights

When employing the EU-SILC data to train the anomaly detection models, it is essential to account for the sample design by means of (adjusted) EU-SILC design weights. As discussed in Section 2.2, the EU-SILC survey adopts a two-stage sampling design in which Comuni serve as PSUs and households as SSUs. Together with the microdata, Eurostat provides design weights calibrated (in the following simply named weights) to available population totals to account for non-response and to enhance estimation efficiency. Weights can be directly applied to training sets and incorporated into the reconstruction or loss function. We then focus on the Horvitz-Thompson estimation, i.e., the re-alignment of data collected under the EU-SILC complex sampling design through inverse probability weighting. In this way, the loss function remains consistent with the underlying survey design, ensuring that the anomaly detection model learns patterns that reflect population-level characteristics. For instance, the reconstruction error in [AEs](#) can be adapted to incorporate survey weights as follows:

$$\tilde{\mathcal{L}}_{\text{AE}}(\mathbf{x}; \theta, \varphi) = \sum_{j=1}^n w_j \|\mathbf{x}_j - g_{\varphi}(f_{\theta}(\mathbf{x}_j))\|^2,$$

where w_j is the weight of the j -th sample and $f_\theta(\cdot)$ and $g_\varphi(\cdot)$ denote the encoder and decoder mappings, respectively. In this formulation, each unit contributes to the reconstruction error in proportion to its survey weight, ensuring that the model emphasises reconstruction fidelity in line with the target population. Similarly, in KPCA, the mapping Φ must be determined from a kernel matrix whose feature space representation has zero mean. Denoting by $\mathbf{K}_{ij}$ the entries of the kernel matrix, we obtain the desired centred kernel matrix $\tilde{\mathbf{K}}_{ij}$ (or Gram Matrix) with incorporated design weights by

$$\tilde{\mathbf{K}}_{ij} = \mathbf{K}_{ij} - \sum_{l=1}^n w_l \mathbf{K}_{il} - \sum_{l=1}^n w_l \mathbf{K}_{lj} + \sum_{m=1}^n \sum_{l=1}^n w_l w_m \mathbf{K}_{lm}.$$

For completeness, the formal notation and derivations are deferred to Appendix B.4.

4 Empirical results on 2019 EU-SILC Liguria data

This section describes the outcome of the methodology from Section 3 applied to the Ligurian subset of the 2019 EU-SILC data.

4.1 Detection of misrepresented subgroups

As introduced in Section 3, the analysis was framed as an outlier detection problem, for which we employed three distinct methodologies. For each method, we present both its effectiveness and the corresponding validation analyses.

The Entropy Score method returns seven atypical variables, reported in Table 1. Five of them are related to specific household characteristics, while the other two pertain to the type of school attended by household members (when applicable), and, more generally, the severity of material deprivation.

For a more detailed investigation, we examined the sampling (marginal) distribution of these variables, as shown in Table 2. Categories with low prevalence ($\leq 10\%$) include individuals who declared they do not hold Italian citizenship ($\sim 5\%$); those who spent a medium- to long-term period abroad (excluding tourism and other leisure-related travel) ($\sim 6\%$); individuals who are not Italian from birth ($\sim 5\%$); those who possess a second citizenship ($\sim 2\%$); and those experiencing material deprivation ($\sim 3\%$). Additional attention is drawn to school attendance ($\sim 8\%$), particularly individuals with no schooling ($\sim 0.08\%$) or enrolled in nursery or kindergarten ($\sim 0.04\%$) and to households in which one or more members did not reside continuously throughout the entire reference year ($< 1\%$).

For the stability validation of the KPCA and the AE methods, we obtained MCC of 0.93 ± 0.01 and 0.94 ± 0.01 , respectively. Additionally, for the internal validation, MCC were equal to 0.98 ± 0.01 and 0.97 ± 0.01 , respectively. Details of the variables inspection are in Tables C3 for KPCA and C4 for AE in Appendix C. In summary, for KPCA the top-5 variables that are most influential in outliers identification are (in decreasing order of importance) the respondent's age, the number of minor children within the household, social exclusion due to poverty, low-work intensity, and the risk of severe material deprivation. For AE, the top-5 variables that are most influential in the outlier

Selected EU-SILC Variables	Entropy Score	Cluster
Lived within the household for the whole 2019	0.951527	Atypical
Secondary Citizenship (if applicable)	0.874917	Atypical
Severity of material deprivation	0.804957	Atypical
Type of school attended (if applicable)	0.737139	Atypical
Italian Citizenship	0.725303	Atypical
Italian citizen from birth	0.714841	Atypical
Continuous residence in Italy	0.657279	Atypical
Risk of poverty indicator	0.498129	Typical
Number of minor Children within the household	0.459679	Typical
Number of women aged 15-55	0.405058	Typical
Social exclusion due to poverty	0.375726	Typical
Source of income or funding	0.288991	Typical
Civil status	0.270583	Typical
Indicator of low work intensity	0.241451	Typical
Number of individuals in the household	0.216393	Typical
Employment status or occupation type	0.143095	Typical
Respondent's age	0.127812	Typical
Type of family	0.117092	Typical
Quintile in the European Union economic ranking	0.026564	Typical

Table 1: Outlier variable detection results using the Entropy Score. The first column provides a description of the 19 EU-SILC variable under consideration. The central column reports the corresponding Entropy Score values. The third column presents the results of a 2-cluster K-means algorithm.

identification are, in decreasing order of importance, the employment status, the main source of funding for the household, the respondent’s age, the number of individuals in the households, and the low work intensity.

To describe profiles of potentially under-represented groups we use medoids. These are reported in full in Tables C5 for KPCA and C6 for AE of Appendix C. Below we summarise our findings. The KPCA analysis identifies the following four subgroups with markedly different socio-demographic profiles

1. minors (0–16) in three-member nuclear families with two parents, showing no material deprivation or poverty.
2. elderly individuals (over 64), often widowed and living alone or in two-member households likely including a caregiver, and fully pension-supported. Born Italian citizens, they show no material deprivation, reflecting a typical post-retirement profile with financial independence but potential social isolation.
3. individuals aged 16–32 in single-parent households with “adult children” only, flagged by poverty risk and social exclusion indicators. Often non-Italian and potentially migrants, they lack stable employment as a primary income source, indicating precarious economic conditions and limited structural support. This subgroup, while demographically younger, is potentially marked by social vulnerability and limited structural support.

Variable Name	Category Frequencies
Continuous residence in Italy	Yes: 0.9361 No: 0.0639 Not applicable: 0
Italian citizen from birth	Yes: 0.9257 No: 0.0473 Not applicable: 0.027
Secondary citizenship	Yes: 0.0171 No: 0.9829 Not applicable: 0
Severity of material deprivation	Yes: 0.0301 No: 0 Not applicable: 0.9699
Type of school attended	Kindergarten: 0.019 Nursery: 0.024 Primary school: 0.036 Lower secondary school: 0.017 Upper secondary school: 0.004 No schooling: 0.007 Not applicable: 0.893
Lived within the household for the whole 2019	Yes: 0.992 No, only for a period: 0.008 No: 0
Italian citizenship	Yes: 0.954 No: 0.047

Table 2: The observed values (frequency) within the categorical variables identified as atypical and potentially indicative of misrepresented subgroups according to the Entropy Score. The left column lists the variable names and the right column reports the relative frequencies of each category.

4. working-age adults (32–64) in dual-adult, child-free households sustained by salaried income, with no signs of deprivation. Relatively stable and middle-income, these households may nonetheless be in transition —possibly with grown children— focused on economic consolidation.

The [AE](#) analysis identifies the following seven subgroup

1. young Italian-born adults in medium-sized households (4 members), economically disadvantaged, relying on cohabiting family support, and flagged for poverty and social exclusion
2. like the previous group, but with more stable economic position and ranking in the top EU income quintile

3. older adults, living in large, extended households, supported by pensions, with no poverty indicators, suggesting a degree of economic stability within a multi-generational living arrangement
4. working-age adults, typically divorced and employed, living in small households, with salaried income and ranked in the middle of the EU income distribution, indicating a profile of economic independence and stability.
5. very large households (9 members) with several children and women of reproductive age, married with dual citizenship and continuous Italian residence, relying on family income support — suggesting a traditional, possibly multicultural structure with complex dependencies
6. small, nuclear family setup with minor children and one working parent, indicating a relatively stable but support-reliant structure
7. household with a mature married couple without children, both employed and in the highest EU income quintile.

4.2 Models’ validation

According to Sections 3.3, all implemented models were subjected to validation procedures: *stability validation* to evaluate sensitivity to single-case perturbations, *internal validation* to assess the model’s consistency and generalisability across the dataset, *cross validation* to evaluate robustness to novel outliers.

For the Entropy Score, the stability validation yielded an **MCC** of 0.94 (95% CI: 0.92–0.96), while the internal validation resulted in an **MCC** of 0.91 (95% CI: 0.89–0.93), and the cross-validation achieved an average **AUROC** of 0.82 (95% CI: 0.79–0.83). For the **KPCA**, we obtained an **MCC** of 0.99 (95% CI: 0.97–1.00) in the stability validation and 0.98 (95% CI: 0.97–0.99) in the internal validation, and the cross-validation achieved an average **AUROC** of 0.83 (95% CI: 0.81–0.85). For the **AE**, both the stability and internal validations achieved an **MCC** of 0.99 (95% CI: 0.98–1.00), and the cross-validation achieved an average **AUROC** of 0.84 (95% CI: 0.82–0.86).

These results confirms the high reliability and internal consistency of the empirical results presented above and a consistent performance across folds.

5 Integrative Sampling Strategies for Enhanced subgroup Representation

The Ligurian EU-SILC dataset provides a case study for detecting misrepresented subpopulations, with particular attention to frail, marginalised, or underserved groups demanding policy attention. Extending beyond standard adjustment procedures which account for sample design, the approach proposed in Section 3 detects and profiles such subgroups by integrating methodological rigour with an ethically informed perspective, aligning with National Academies of Sciences and Medicine [17]. The resulting evidence motivated, within the Liguria Project (Section 2.1), the consideration of a supplementary survey targeting households identified as vulnerable through Tax Declaration Data linkage (Section 2.3.1) and the analysis in Section 4. This section explores possible designs for such a complementary survey via a simulation exercise, offering

both methodological insights and practical recommendations for enriching EU-SILC data and supporting regional policymaking.

5.1 Simulated Sampling Strategies

Two sampling strategies are considered, each pairing a sample design with an estimator for the target population quantity. They differ in their data and contextual requirements, but share the aim of controlling sample size for targeted subgroups while accounting for varying availability of sampling frames and population-level para-data. Both strategies also encompass, as special cases, sampling campaigns targeting a single misrepresented subgroup.

The first strategy uses a stratified design, applicable when two conditions hold: (1) high-quality regional auxiliary data (e.g., from a census) are available to define strata and boost representation of targeted subgroups; and (2) a complete regional population list, including all subgroups of interest for the supplementary survey, exists and can serve as a reliable sampling frame for both stratification and sample selection, following e.g. Barron et al. [3]. When met, standard unbiased estimation and allocation methods can handle budgetary and accuracy constraints. In practice, however, these conditions may be hard to satisfy. A practical alternative relies on two or more partial sampling frames, pre-existing or purposely constructed, with high expected prevalence of the target subgroups, leading to the second strategy.

The second strategy is a Multiple Frame Survey (MFS) using $Q \geq 2$ partial frames, independently sampled, with data combined to produce target estimates. It applies when: (1) the union of the Q frames adequately covers the targeted subgroups; and (2) frames may overlap, so units appearing in multiple frames have higher selection probabilities (a issue known as *multiplicity*). Multiplicity is addressed via a Multiple Frame estimator with multiplicity-adjusted weights, requiring frame membership data collected during the survey. Following the MFS literature, two estimators are considered: the *Simple Multiplicity* (SM) estimator (see Mecatti [14], Mecatti and Singh [24]) and the *Pseudo Maximum Likelihood* (PML) estimator (see Skinner and Rao [21], Lohr and Rao [23]). The SM estimator is unbiased and operationally simple, requiring only basic frame membership information for a fixed multiplicity correction, though potentially inefficient. The PML estimator is nearly unbiased and typically more efficient, but more complex and requiring more detailed membership data.

5.2 Simulative Scenarios

To implement the proposed sampling strategies and assess their relative effectiveness, a simulation study was conducted. A synthetic population of size $N = 3000$ was generated from a bivariate random variable (X, Y) with a Gaussian distribution with parameters $\mu_x = 4$, $\mu_y = 1$, $\sigma_x = 0.3$, $\sigma_y = 0.2$, and $\rho_{xy} = 0.85$. In order to reflect naturally bounded socio-economic variables, the distribution was truncated at zero to ensure positivity of the population values for both variables. In this setting, Y represents the study variable, with the population mean μ_y as the parameter to be estimated, and X serves as the design variable. For the stratified sampling strategy (hereafter denoted as STS), the population was partitioned into $Q = 3$ strata based on

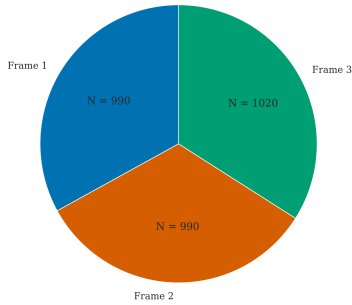
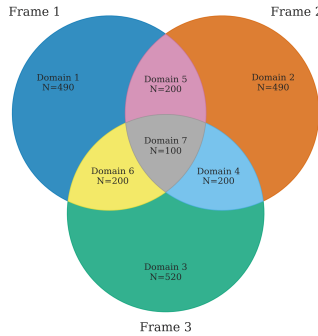
Population				
Stratified Sampling 		Multi-Frame 		
Sample				
Design	Frame	Allocation	Sampling Fraction	Sample Size
Stratified (STS)	Frame 1	Proportional	0.20	198
Stratified (STS)	Frame 2	Proportional	0.20	198
Stratified (STS)	Frame 3	Proportional	0.20	204
Stratified (STS)	Frame 1	Optimal Cost	0.35	213
Stratified (STS)	Frame 2	Optimal Cost	0.28	171
Stratified (STS)	Frame 3	Optimal Cost	0.36	217
Multi-Frame (MF)	Frame 1	Proportional	0.20	198
Multi-Frame (MF)	Frame 2	Proportional	0.20	198
Multi-Frame (MF)	Frame 3	Proportional	0.20	204
Multi-Frame (MF)	Frame 1	Optimal Cost	0.26	156
Multi-Frame (MF)	Frame 2	Optimal Cost	0.41	246
Multi-Frame (MF)	Frame 3	Optimal Cost	0.33	198

Table 3: Population and Sample Allocation under Stratified and Multi-Frame Sampling Designs

the variable X . Specifically, three strata were constructed based on the 33rd and 66th percentiles of X , thereby producing approximately equal-sized strata in a practical and efficient manner. For the MFS, the N population units were randomly assigned to $Q = 3$ overlapping frames, with sizes reported in Table 3. For each strategy, the total sample size was fixed at $n = 600$ (corresponding to a sampling fraction of $n/N = 0.2$), and two sample allocation scenarios across the three strata or frames were considered:

1. a *proportional allocation*, with the same sampling fraction (0.2) applied to each stratum or frame, and
2. an *optimal allocation* under a cost constraint defined by a linear cost function:

$$C = c_0 + \sum_{q=1}^Q n_q c_q,$$

where n_q denotes the sample size allocated to stratum (or frame) q . The fixed cost was set to $c_0 = 0$, while the per-unit survey costs were set to $c_1 = 9.5$, $c_2 = 10$, and $c_3 = 10.5$.

A simple random sample was selected within each stratum for the STS, whereas a Rao-Sampford sample design, introduced in Sampford [19], was used within each frame for the MFS. For each strategy and each allocation scenario, $M = 2500$ samples were generated, producing simulated distributions of the three estimators for μ_y discussed in Subsection 5.1. The relative performance of the sampling strategies was evaluated using standard Monte Carlo (MC) metrics, namely the MC mean squared error (MSE) and relative bias (RB) as given by:

$$\text{MSE}_{MC}(\hat{\mu}_y) = \frac{1}{M} \sum_{m=1}^M (\hat{\mu}_{y,m} - \mu_y)^2, \quad \text{RB}_{MC}(\hat{\mu}_y) = \frac{1}{M} \sum_{m=1}^M \frac{\hat{\mu}_{y,m} - \mu_y}{\mu_y}.$$

where $\hat{\mu}_{y,m}$ denotes the estimate obtained at the m -th MC iteration for the estimator under consideration. Further details on the simulation setup and the sample allocation procedures are provided in Appendix D.

5.3 Monte Carlo Results

Figure 2 illustrates the convergence of the simulated estimators ([Stratified Sampling \(STS\)](#), [Simple Multiplicity \(SM\)](#), and [Pseudo-Maximum Likelihood \(PML\)](#)) across iterations in the MC experiment. Based on these results, 2500 MC iterations are sufficient for the simulation setting considered, as at this point the interval defined by P_5 and P_{95} , i.e. the 5th and 95th percentiles of the MC empirical distributions, contains the true population mean, and the relative error between the estimated MC mean and the true mean is effectively below 1%. Results in Table 4 for the MC metric MSE_{MC} are consistent with the data and context requirements discussed in Subsection 5.1 for the actual implementation of the two sampling strategies simulated. For the [STS](#),

Allocation	Estimator	Mean Square Error (MSE)	P_5	P_{95}
optimal_cost	PML	26.5	23.3	29.8
optimal_cost	SM	27.1	23.8	30.4
optimal_cost	STS	24.9	21.6	28.2
proportional	PML	26.7	23.4	30.1
proportional	SM	27.5	24.1	30.6
proportional	STS	26.2	22.9	27.5

Table 4: omparison of mean [MSE](#) and corresponding 5th and 95th percentiles under different allocation strategies and estimators. Values are reported in units of $\times 10^{-6}$

optimal-cost allocation shows a slight 5% advantage over proportional allocation. In the multi-frame setting, the [PML](#) estimator is more efficient under both allocation

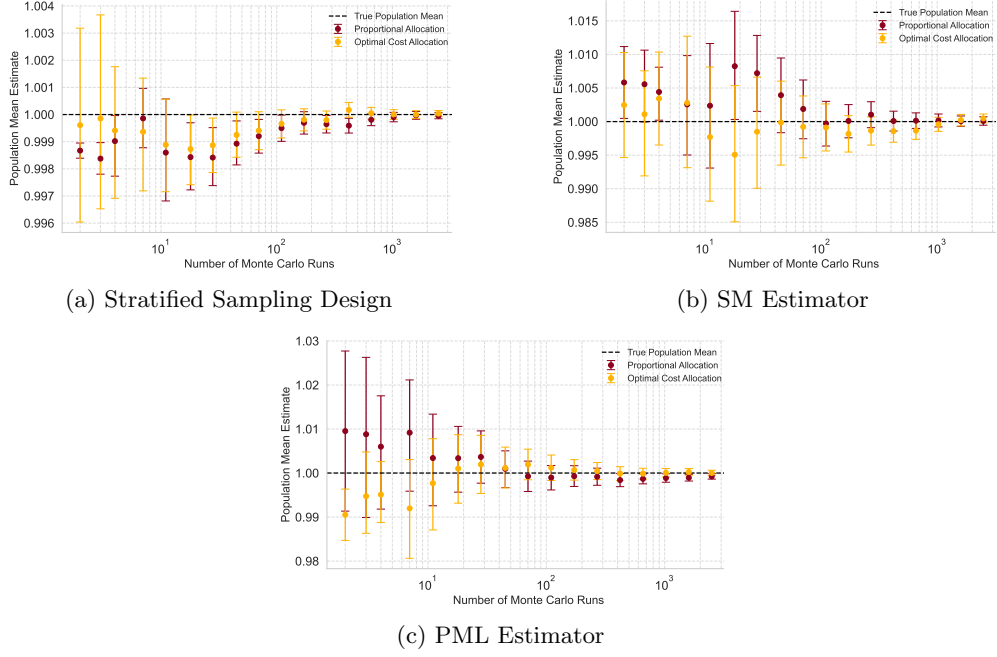


Fig. 2: Estimated population means across increasing Monte Carlo replications under (a) stratified sampling, (b) MF sampling with SM, and (c) MF sampling with PML, shown for both proportional and optimal cost allocation. Trajectories illustrate convergence toward the theoretical mean (dotted line); error bars denote 95% bootstrap confidence intervals.

strategies; however, the **SM** performs comparably and, being simpler to compute with less demanding data requirements, is a practical and recommendable alternative.

For the RB_{MC} (Figure 3), no substantial differences across allocation strategies are observed, with all boxplots centred near zero, consistent with the theoretical unbiasedness of **STS** and **SM** and quasi-unbiasedness of **PML**. In the **STS** case (Figure 3a), **Relative Bias (RB)** remains low (between -0.01 and 0.01) with a slight positive tendency. In the multi-frame setting, distributions are more symmetrically centred around zero but show greater variability (approximately -0.05 to 0.05), with both **SM** and **PML** exhibiting lower dispersion and confirming their stability.

These results highlight that sampling strategy selection should be context-driven. When a complete frame is available and stratification is feasible, stratified sampling offers a straightforward solution. When only partial frames are accessible, particularly those containing targeted subgroups for oversampling, an MFS strategy is recommended. Among multi-frame estimators, **PML** achieves superior efficiency but requires greater para-data and computational effort; in such cases, **SM** is a viable alternative, combining theoretical unbiasedness with simpler implementation. Additionally, multi-frame designs naturally accommodate differential frame costs, where cost-sensitive allocation can yield efficiency gains over standard single-frame approaches.

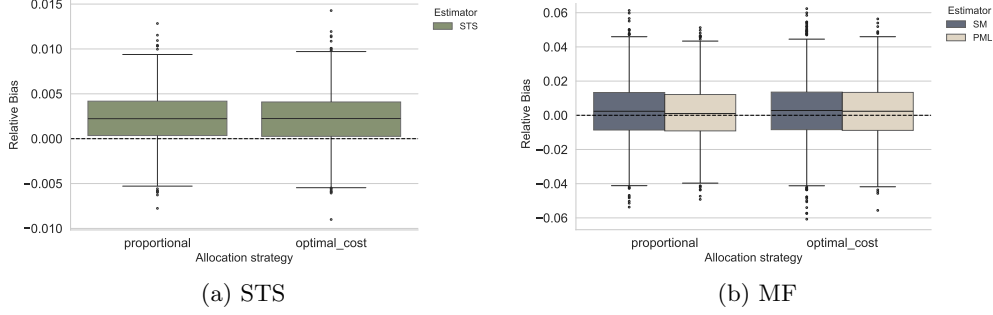


Fig. 3: Box plot for the RB estimation for STS and MF estimators across Monte Carlo iterations per allocation strategy

6 Discussion and conclusion

As outlined in Section 1, this work proposes a framework for identifying misrepresented subgroups in survey datasets such as EU-SILC. Although EU-SILC’s design weights ensure unbiased estimation and valid inference at national and sub-national level, certain subgroups may remain insufficiently represented, limiting the dataset’s reliability for socio-demographic policy targeting at national or regional level. The proposed methodology can be interpreted within a design-based framework, focused on the identification and integration of informative units associated with misrepresented subpopulations. From this perspective, the problem can be viewed as a data-driven design task, where the goal is to detect regions of the population space that are insufficiently represented in the original sample and to guide subsequent sampling efforts accordingly.

Misrepresented subgroups often coincide with hard-to-reach populations. Detecting those subgroups and under-coverage typically requires prior knowledge of the subpopulations of interest and a reliable external reference dataset, serving as a ground truth but rarely available in practice. Given this limitation, we reformulate the problem as outlier detection, assuming that atypical data patterns reflect either hard-to-reach groups or misrepresented subpopulations deviating from the population core. The idea is that hard-to-reach populations, due to their inherent difficulty in being surveyed, tend to exhibit atypical patterns and thus are likely to emerge as outliers. Thus, groups significantly deviating from the “core” characteristics of the surveyed population may be indicative of inadequate representation of specific subgroups.

Our framework combines univariate and multivariate methods, including kernel-based approaches and an auto-encoder (AE), not as redundancy, but to assess whether subgroups are consistently detected under increasing model flexibility. Convergence across heterogeneous methods strengthens the credibility of findings in the absence of ground truth. Applied to EU-SILC data for Liguria, our findings reveal specific misrepresented household types. Entropy-based analysis identified gaps related to non-continuous residency, non-Italian citizenship, school-age children, and material deprivation. The AE confirmed these patterns, detecting economically vulnerable

youth and contrasting them with affluent or stable households, while prominently identifying large, multicultural, multi-family households often overlooked by standard surveys. KPCA confirmed vulnerability and affluence patterns, and, crucially, revealed misrepresented subpopulations defined by joint feature combinations undetectable through univariate analysis alone. These findings lay the groundwork for targeted complementary data collection or integration. The methodology is readily transferable beyond EU-SILC to any setting where subgroup misrepresentation is suspected.

The proposed framework consists of two components with complementary roles: the detection phase and design phase. The sampling phase is presented through a controlled simulation study, which illustrates how the identified misrepresented subgroup populations could be more effectively targeted and investigated in subsequent data collection. A comparative simulation study showed that no single approach is universally optimal: when a complete frame is available, **STS** is recommended for its simplicity; when only partial frames exist, a multi-frame strategy with **PML** is preferred, despite its greater implementation demands compared to **SM**. The main limitation is the absence of ground truth, making external validation impossible and rendering findings inherently data-specific. Nevertheless, internal validation, sensitivity analyses, and a cross-validation procedure provide a proof of concept, confirming that the methods consistently capture similar features across runs.

Data and Code Availability

The data used in this study are drawn from the EU-SILC survey and are subject to access restrictions; they are available upon request from the relevant data providers. Python code available at https://github.com/glancia93/Data_Driven_Strategies_for_Detecting_and_Sampling_misrepresented_subgroups

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Appendix A Supplementary Background, Context, and Data

A.1 Social background

Despite the proliferation of new technologies, poverty and inequality persist in European societies, and the anticipated positive effects of technological advancement on these socioeconomic issues have not materialised as expected Atkinson [2], Longford [22]. In opposition to this, the role of education in contrasting poverty and inequality has been largely proven Hofmarcher [17], Raffo et al. [25]. Among countries in the EU-SILC programme, low-level education has been correlated with a gap in income that affects children’s educational outcomes Ferguson et al. [14]. In addition, integration plays an important role in comprehending social inequalities, such as the impact of unskilled immigrants on natives Gang and Rivera-Batiz [15] and the difficulty these immigrants experience in owning a home or earning higher incomes once migrated with few resources Lewin [20].

The Ligurian *Pact for Residential Stability* and *Family Support* measures were designed to reduce the fiscal burden on households and stimulate investment and consumption; see art. 5 of Regione Liguria [26]. The policies were proposed in 2017, approved in 2018, and introduced in 2019. Specifically, for that year, tax exemptions were granted to large families and citizens with a low or medium-low taxable income, provided they had at least one dependent child born in 2018 Regione Liguria [26]. In order to develop an efficient rich plan of assistance services and extend them to those groups of the resident population that are more exposed to inequality and poverty factors, referred to as vulnerable, frail and marginalised residents, the administrative bureau of the Liguria region has recently proposed to fill the gap in the available data, particular in the Ligurian sub-sample of the EU-SILC programme, on their specific challenges. Our work has the ultimate scope to address this gap.

A.2 2019 EU-SILC survey in Italy

The main purpose of the EU-SILC programme is to collect timely and comparable cross-sectional and longitudinal multidimensional micro-data on income, poverty, social exclusion and living conditions, produced annually and covering all 27 EU Countries plus Iceland and Norway Eurostat [8, 9].

ISTAT [18] (the Italian official statistics agency) in its 2019 EU-SILC methodological report guarantees a sample size of (circa) 22000 households and 50000 individuals across 650 municipalities (Comuni), leading to a national sample fraction of 8.4 per 10000 individuals (0.084%).

In addition to the EU general guidelines requests, ISTAT also assures representative (cross-sectional) probabilistic samples and precise estimates at the regional level. No minimum effective sample size is prescribed for the Ligurian cross-sectional sample, as sampling is conducted at the municipality level rather than at the regional level. Specifically, the 2019 Ligurian cross-sectional sample comprises 1925 individuals from 990 households across 21 municipalities, corresponding to an individual sample fraction of 0.3 per 10,000 (0.003%) of the Italian population and 13 per 10,000 (0.13%) of

the Ligurian population. The EU-SILC Italian sample design has two stages with stratification of the primary sample units (PSU) represented by Comuni and secondary sample units (SSU) represented by households. The first stage sample is selected with probability proportional to PSU demographic size and includes: the largest PSUs that inform a unique stratum, four PSUs selected within strata informed by at least four Comuni; and, otherwise, two selected PSUs. The second stage sample is a simple random without replacement (SRS).

According to ISTAT [18] collected data are processed and edited; missing data imputed, also on the basis of external data from administrative official registries; original design weights, which are uniquely defined as the inverse of the (first order) inclusion probability, are adjusted for item non-response and according to two steps of calibration to available known total controls. Access to aggregated data is open to the general public. Microdata can be accessed on demand for scientific purposes and projects (<https://ec.europa.eu/eurostat/web/microdata/european-union-statistics-on-income-and-living-conditions>).

A.3 Selected EU-SILC Variables

EU-SILC Acronym	Variable's Description
ETAINT	Age group (16-32, 32-64, 65+ years)
CITTADX	Italian Citizenship
NCITT	Italian citizen from birth
SECITT	Secondary Citizenship (if applicable)
ITA	Continuous residence in Italy
VIFAM	Lived within the household for the whole 2019
LAVPRI	Employment status
FONTERED	Source of income or funding
TOT_TUTTI	Number of individuals in the household
RAGA017	Number of minor Children within the household
DON1555	Number of women aged 15-55
STACIV	Civil status
TF	Type of family
TIPSCU	Type of school attended (if applicable)
SEV_MAT_DEPRIV	Severity of material deprivation
RISKPOV	Risk of poverty indicator
LOW_WORK_INT	Indicator of low work intensity
POV_SOC_EXCL	Social exclusion due to poverty
QUINTLEU	Quintile in the European Union economic ranking

Table A1: List with the description of the 19 categorical variables selected for the analysis.

Table A1 shows a summarising description of the variables extracted from the EU-SILC dataset and their description. The *severe material deprivation*

(EV_MAT_DEPRIV) variable reflects the condition in which a person cannot afford at least four out of nine essential items. Let N_ITEM be the number of items a person cannot afford, with values ranging from 0 to 9. Those items are

1. to pay their rent, mortgage or utility bills;
2. to keep their home adequately warm;
3. to face unexpected expenses;
4. to eat meat or proteins regularly;
5. to go on holiday;
6. a television set;
7. a washing machine;
8. a car;
9. a telephone.

The severe material deprivation rate, broken down by subgroup k (denoted SEV_DEPR_TOTALat_k), is calculated using the adjusted weight RB050a as:

$$\text{SEV_DEPR_TOTALat_k} = \frac{\sum_{i \in k, N_ITEM \geq 4} \text{RB050a}_i}{\sum_{i \in k} \text{RB050a}_i} \times 100.$$

The threshold of four items to depict severe material deprivation has been chosen for a mixture of empirical and practical reasons since a previous threshold of 3 items had resulted in excessively high, and politically unmanageable, estimates of levels of deprivation across the EU. This gives the percentage of individuals in group k experiencing severe deprivation; see Eurostat [12].

The *at-risk-of-poverty rate* (RISKPOV) indicates the proportion of individuals whose disposable income, adjusted for household size (equivalised), falls below 60% of the national median after accounting for social transfers. This measure reflects income inequality rather than absolute poverty, highlighting how someone fares economically relative to the rest of the population in their country. That is, this indicator does not necessarily imply poor living conditions but rather signals a lower income compared to the general population; see Eurostat [11].

The indicator *people living in households with very low work intensity* (LOW_WORK_INT) captures the share of individuals aged 0–64 who live in households where the working-age members (18–64 years old) have, on average, worked less than 20% of their full work-time potential during the previous year. Work intensity is calculated as the ratio between the number of months actually worked by all working-age adults in the household and the total number of months they could theoretically have worked. These months are counted in full-time equivalent terms: part-time work is adjusted based on the number of usual working hours reported during the interview. To ensure consistency, the calculation excludes individuals who are students aged 18–24, retirees (based on their self-declared economic status or pension receipt), and people aged 60–64 who are inactive and live in pension-dependent households. Households made up only of children, students under 25, or elderly people (65+) are also excluded from this indicator; see Eurostat [13].

The indicator *At Risk of Poverty or Social Exclusion* (AROPE) measures the share of the population facing at least one of the following three conditions: risk of poverty, severe material and social deprivation, or living in a household with very low work intensity (from the variable POV_SOC_EXCL). Each person is counted only once, even if they fall into multiple categories. AROPE is the main metric used to track progress toward the European Union’s 2030 poverty and social exclusion targets, and it also serves as the headline indicator for the EU 2020 Strategy Eurostat [10].

Appendix B Mathematical Details on Anomaly Detection Methods

B.1 Entropy Score

The *Shannon entropy* (or simply, entropy) of a discrete random variable X , assuming values over a sample space $\mathcal{X}$ and distributed according to a probability mass function P , is defined as

$$H[P] = -\mathbf{E}_{X \sim P}[\log P(X)]. \quad (\text{B1})$$

By construction, $H[P] \geq 0$ and is a concave functional Bishop and Nasrabadi [5]. For a discrete distribution with N categories, entropy is bounded between 0 and $\log N$. The maximum value $\log N$ occurs when all categories are equally likely, while the minimum value of zero corresponds to a degenerate distribution where a single category holds all the probability mass.

We used entropy in Section 3.1.1 as a scoring criterion to build an outlier model for identifying the most anomalous variables in the EU-SILC dataset. In the one-dimensional discrete setting, increasing entropy reflects a progressive *spreading* of the probability mass across categories. Thus, higher entropy corresponds to a more uniform distribution, whereas lower entropy signals a concentration of probability on a few categories. This relationship between entropy and distributional spread is central to our sampling perspective.

Entropy and other information theory-based measures have long been used in anomaly and outlier detection, for example by Chandola et al. [6]. Due to entropy’s ability to quantify the informativeness of a source, it has also been proposed as a mechanism for automated selection of anomalous samples by Berezinski et al. [4], Daneshpazhouh and Sami [7]. We adapted entropy’s properties to a univariate analysis framework, under the simplifying assumption that the dataset variables can be considered temporarily independent. Such a simplification allows the capture *independent patterns* that most strongly characterise misrepresented subgroups. The method we propose is twofold: first, we consider a variable-specific entropy-based score to identify the most informative variables, those most likely to present outliers; second, we exploit the notion of *information* associated with entropy to determine which categories within those variables are most representative of the target outliers.

Formally, let V be a univariate categorical variable taking values in the set of categories $\{C_1, \dots, C_N\}$, with observed data $x_1, \dots, x_n$. The entropy-based score we

propose is the following:

$$\Gamma_1 = 1 + \frac{1}{\log N} \sum_{k=1}^N p_k \log p_k, \quad (\text{B2})$$

where p_k denotes the empirical probability of category C_k , given by:

$$p_k = \frac{1}{n} \sum_{j=1}^n \mathbf{1}(x_j \in C_k). \quad (\text{B3})$$

Γ_1 may take values only in $[0, 1]$; the higher the value of Γ_1 , the more the possibility of finding under-represented categories. Once the values of Γ_1 have been analysed to identify the top five most informative variables (see Section 3.2), we examine the information content of each variable, quantified as $-\log p_k$, to determine which group is misrepresented within each selected variable.

B.2 KPCA

Let $\hat{X} \in \mathbb{R}^{n \times p}$ be the data matrix, where n denotes the number of sampled units and p the number of variables. Denote by $\mathcal{X} \subseteq \mathbb{R}^p$ the input space of the observations (i.e., the space to which each row vector $\mathbf{x} \in \mathbb{R}^p$ belongs). **KPCA** assumes the existence of a (possibly nonlinear) feature map $\Phi : \mathcal{X} \rightarrow \mathcal{H}$, where $\mathcal{H}$ is a (typically high-dimensional) Hilbert space, referred to as the feature space. This mapping is defined such that the inner product between any two mapped observations can be computed via a positive semidefinite kernel function:

$$k(x, y) = \Phi^T(\mathbf{x})\Phi(\mathbf{y}) \quad \mathbf{x}, \mathbf{y} \in \mathcal{X}.$$

Therefore, **KPCA** treats the data as if they are mapped into a richer space where their structure becomes easier to separate. The kernel function enables the computation of how similar two unites are in this space without explicitly performing the mapping.

The transformed observations $\Phi(\mathbf{x})$ are never computed explicitly. Instead, the analysis is carried out through the *Gram matrix* $\mathbf{K} \in \mathbb{R}^{n \times n}$, whose entries are defined as

$$K_{xy} = k(\mathbf{x}, \mathbf{y}).$$

Since **PCA** requires the data to be centred, the same operation must be performed in the feature space induced by Φ . Because this feature space is not explicitly available, centring is achieved by centring the Gram matrix itself. This operation produces the same effect as subtracting the mean of the mapped data in the (implicit) feature space. In essence, centring the Gram matrix ensures that **KPCA** is conducted on mean-centred data, even though the feature map Φ is never computed directly.

After centering $\mathbf{K}$ to obtain the *centered Gram matrix* $\tilde{\mathbf{K}}$, **KPCA** performs eigen-decomposition by solving:

$$\tilde{\mathbf{K}}\mathbf{v} = \lambda\mathbf{v},$$

where $\mathbf{v}$ are the eigenvectors corresponding to the principal directions in the feature space, and λ are the associated eigenvalues. In essence, **KPCA** applies **PCA** to the non-linearly transformed data $\Phi(\mathbf{x})$, enabling the capture of complex, non-linear structures in the original space. Unlike standard **PCA**, the eigenvalues λ no longer represent variances in the original input space, but instead quantify the variance of the data as expressed in the embedding space $\mathcal{H}$.

For outlier detection, an early application of **KPCA** explored the geometry of the transformed feature space induced by the *radial kernel*, as proposed in Hoffmann [16]. In this framework, *typical observations* are expected to lie close to the principal subspace $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$, spanned by the top k eigenvectors whose associated eigenvalues explain a significant portion (we opted for 99%) of the variance in the embedding space $\mathcal{H}$.

Given a new input $\mathbf{z} \in \mathbb{R}^p$, its image under the feature map is $\Phi(\mathbf{z}) \in \mathcal{H}$. The projection of $\Phi(\mathbf{z})$ onto the subspace defined by the top k principal components is then:

$$\hat{\Phi}(\mathbf{z}) = \sum_{i=1}^k [\mathbf{v}_i^T \Phi(\mathbf{z})] \mathbf{v}_i,$$

and the *reconstruction error*, which serves as an unsupervised outlier score, is defined as:

$$\text{RE}_{\text{KPCA}}(\Phi(\mathbf{z})) = \|\Phi(\mathbf{z}) - \hat{\Phi}(\mathbf{z})\|^2 = \sum_{i=k+1}^p |\mathbf{v}_i^T \Phi(\mathbf{z})|^2.$$

To apply **KPCA** in our context, which involves ordinal and categorical variables, we adopt a kernel based on the *Hamming distance*. We recall that the Hamming distance is given by:

$$\text{Ham}(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^p \mathbf{1}\{\mathbf{x}_i \neq \mathbf{y}_i\}.$$

Thus, the kernel we propose is:

$$k(\mathbf{x}, \mathbf{y}) = \exp(-\gamma \cdot \text{Ham}(\mathbf{x}, \mathbf{y})),$$

where $\gamma > 0$ is a bandwidth parameter controlling the rate of decay; for convenience, we set $\gamma = 1$.

This kernel function is positive semidefinite, a property required to ensure that the kernel defines a valid inner product in $\mathcal{H}$; see Shawe-Taylor and Cristianini [29]. More generally, we recall that metric distances, such as the Hamming or the Euclidean distance, can be transformed into valid kernels by applying the exponential function to distances themselves with the sign changed; see Bekka et al. [3], Schoenberg [27].

B.3 Auto-Encoder and Neural Network

Unlike other reconstruction-based methods, neural networks have the advantage of being universal approximators of any measurable function which establishes the superiority of **AE**-based methods in exploring optimal embedding transformations to make unsupervised learning efficient (see e.g. Aggarwal [1], Bishop and Nasrabadi [5]).

Formally, let us consider a data matrix, $X \in \mathbb{R}^{n \times p}$, where n denotes the number of samples (observations), p denotes the number of features (variables) measured for each sample and the (i, j) element of X is the value of the feature j observed for the i -th sample. A generic row of X is indicated by $\mathbf{x}$. An encoder is defined as a map $f_\theta : \mathbb{R}^p \rightarrow \mathbb{R}^d$ that maps the input $\mathbf{x} \in \mathbb{R}^p$ to a latent representation $\mathbf{z} \in \mathbb{R}^d$; usually $d < p$, and so $\mathbf{z}$ contains a smaller, simplified version of the information in $\mathbf{x}$. Likewise, a *decoder* is a map $g_\varphi : \mathbb{R}^d \rightarrow \mathbb{R}^p$ that reconstructs the input from the latent representation.

The subscripts θ and φ denote the sets of parameters learned in the hidden layers of the encoder and decoder, respectively. Such a model is trained to minimise the reconstruction error, which represents how the compression-decompression mechanism deviates from the input data, considered here as a benchmark. Put simply, the AE learns a sort of optimal transform along with its inverse, to reduce the dimensionality of input data; for this reason, this method represents a wide use adaptive strategy for data reduction in unsupervised learning. Typically, the reconstruction error is expressed as:

$$\mathcal{L}_{\text{AE}}(\mathbf{x}; \theta, \varphi) = \frac{1}{n} \sum_{j=1}^n \|\mathbf{x}_j - g_\varphi(f_\theta(\mathbf{x}_j))\|^2.$$

B.4 Integration of Design Weights in Anomaly Detection

In Section 3.5 we discussed how to generalise any of the anomaly detection methods, and include sample weights in model training phases. In practice, the weights can be directly applied to the training data and incorporated into the loss function. Thus, the model learns patterns that are representative of the target population, rather than being biased by the sample design. For a generic variable X , the weighted mean over a sample of size n is simply

$$\bar{x} = \sum_{j=1}^n w_j x_j, \quad (\text{B4})$$

For instance, in the case of the *Entropy Score*, which relies on histogram-based estimates, frequencies must be weighted accordingly. The weighted formulation becomes:

$$\begin{cases} \tilde{\Gamma}_1 = -\frac{1}{\log N} \sum_{k=1}^N \tilde{p}_k \log \tilde{p}_k, \\ \tilde{p}_k = \frac{1}{n} \sum_{j=1}^n w_j \mathbf{1}(x_j \in C_k); \end{cases}$$

where w_j is the weight of the j sampled unit; C_k denotes the k -th category

For KPCA, the eigen decomposition is carried out on the centred Gram matrix $\tilde{\mathbf{K}}$ obtained once the contribution of the mean feature is removed in the embedding space $\mathcal{H}$, that is,

$$\tilde{\mathbf{K}}_{xy} = \tilde{\Phi}^T(\mathbf{x}) \tilde{\Phi}(\mathbf{y}), \quad \text{with} \quad \sum_{\mathbf{x}} \tilde{\Phi}(\mathbf{x}) = 0,$$

where the centred feature map is defined as

$$\tilde{\Phi}(\mathbf{x}) = \Phi(\mathbf{x}) - \mu_{\mathbf{x}}.$$

where $\mu_{\mathbf{x}}$ denotes the mean effect of the feature $\mathbf{x}$ under the distribution of samples. In the weighted setting, this mean depends on both the feature $\mathbf{x}$ and the sample weights $\mathbf{w} = (w_1, \dots, w_n)^\top$, such that

$$\mu = \sum_{j=1}^n w_j \Phi(\mathbf{x}_j),$$

where $\mathbf{w}$ are the same normalised weights used in Equation (B4). Substituting this definition of the mean into the kernel expression yields the following weighted centring formula for the entries of the Gram matrix:

$$\tilde{\mathbf{K}}_{ij} = \mathbf{K}_{ij} - \sum_{l=1}^n w_l \mathbf{K}_{il} - \sum_{l=1}^n w_l \mathbf{K}_{lj} + \sum_{l=1}^n \sum_{m=1}^n w_l w_m \mathbf{K}_{lm}. \quad (\text{B5})$$

The four terms in (B5) have an intuitive interpretation: the first is the raw similarity $\mathbf{K}_{ij}$, the next two subtract the contributions of the mean from each argument, and the final term adds back the mean–mean similarity. Therefore, applying (B5) directly to the unweighted, uncentered Gram matrix is sufficient to obtain the correctly centred version. The eigen-decomposition of $\tilde{\mathbf{K}}$ provides the principal components in the feature space. If $\tilde{\mathbf{K}}\boldsymbol{\alpha} = \lambda\boldsymbol{\alpha}$, then the associated principal axis in $\mathcal{H}$ is $\mathbf{v} = \sum_{i=1}^n \alpha_i \tilde{\Phi}(\mathbf{x}_i)$, and the projection of a new sample $\mathbf{X}$ onto $\mathbf{v}$ is obtained via the centred kernel

$$\tilde{k}(\mathbf{x}_i, \mathbf{x}) = k(\mathbf{x}_i, \mathbf{X}) - \sum_{l=1}^n w_l k(\mathbf{x}_l, \mathbf{X}) - \sum_{l=1}^n w_l k(\mathbf{x}_i, \mathbf{x}_l) + \sum_{l=1}^n \sum_{m=1}^n w_l w_m k(\mathbf{x}_l, \mathbf{x}_m).$$

For the AE model, the adjustment of the loss function follows directly from the observation that the reconstruction error is given by the MSE between the original input $\mathbf{x}$ and its reconstruction obtained via the composition of the encoder f_θ and decoder g_φ . Explicitly, the standard loss takes the form

$$\mathcal{L}_{\text{AE}}(\mathbf{x}; \theta, \varphi) = \frac{1}{n} \sum_{j=1}^n \|\mathbf{x}_j - g_\varphi(f_\theta(\mathbf{x}_j))\|^2.$$

Since the MSE corresponds to an average, it is natural and legitimate to apply the adjusted averaging scheme introduced in (B4). Incorporating the sample weights w_j , the weighted auto-encoder loss becomes

$$\tilde{\mathcal{L}}_{\text{AE}}(\mathbf{x}; \theta, \phi) = \frac{1}{n} \sum_{j=1}^n w_j \|\mathbf{x}_j - g_\phi(f_\theta(\mathbf{x}_j))\|^2, \quad (\text{B6})$$

This adjustment ensures that the optimisation process accounts explicitly for the sampling design, thereby aligning the training objective with the target population distribution. As a result, the learned representations and reconstructions are not

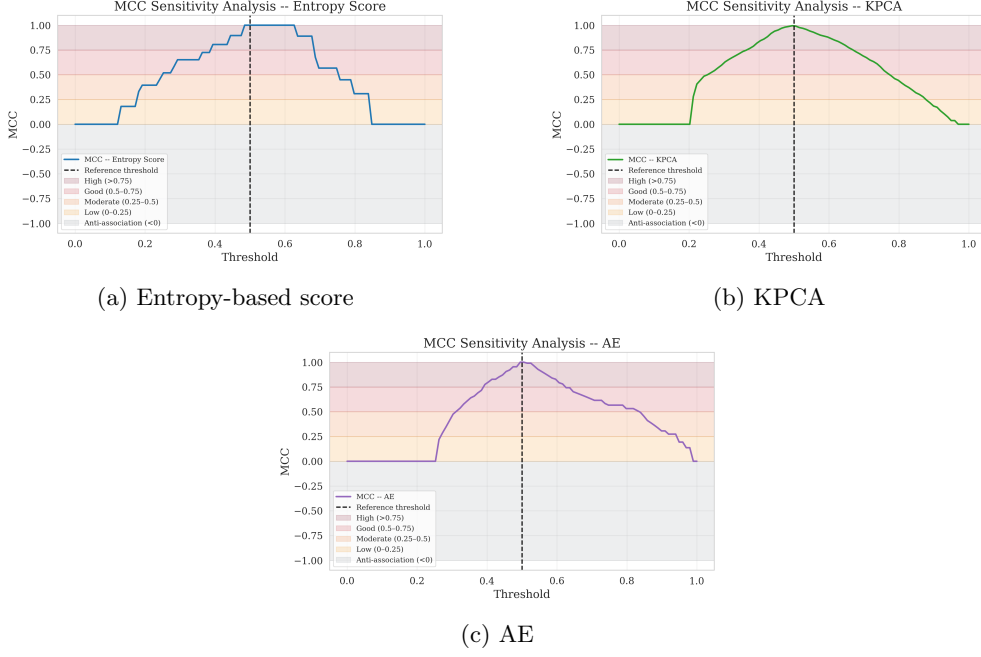


Fig. B1: Sensitivity of outlier detection with respect to threshold variation. The vertical reference line indicates the threshold derived via k -means clustering.

only optimise for the observed sample, but are also valid and representative for population-level inference.

B.5 Sensitivity Analysis of Threshold Selection

To assess the robustness of the proposed thresholding strategy, we conducted a sensitivity analysis focusing on the stability of the outlier identification process with respect to variations in the threshold. In particular, we evaluated whether the partition induced by the k -means-based threshold (with $k = 2$) remains consistent under alternative thresholding schemes and small perturbations of the score distribution. First, we compared the k -means-derived threshold with a set of benchmark thresholds. Without loss of generality, the outlier scores were rescaled to the interval $[0, 1]$ using a sigmoidal transformation. Under this derived transformation, the k -means-derived threshold is centered around 0.5, allowing the exploration of the full range of possible thresholds in a standardised manner. By sweeping the entire $[0, 1]$ domain, we computed the corresponding sets of detected outliers and quantified their agreement with the k -means partition using the [MCC](#). The results show a high degree of consistency across a wide range of thresholds, with the k -means-based solution aligning closely with the region of maximum [MCC](#). This indicates that the proposed threshold is not only data-driven but also near-optimal in terms of agreement with alternative thresholding schemes (see Figure [B1](#)). In addition, we further investigated the adequacy

Model	1-component GMM (AIC)	2-component GMM (AIC)
Entropy-Based Score	2.73	-2.12
KPCA	-456.21	-2156.45
AE	-4201.74	-9707.54

Table B2: Comparison of Gaussian Mixture Models with different numbers of components based on AIC.

Selected EU-SILC Variables	Average Importance (KPCA)
Respondent's age	0.728312
Number of minor children within the household	0.727142
Social exclusion due to poverty	0.726549
Indicator of low work intensity	0.725653
Risk of poverty indicator	0.724115
Severity of material deprivation	0.723654
Type of family	0.722987
Civil status	0.721763
Number of women aged 15-55	0.720985
Number of individuals in the household	0.719878
Italian Citizenship	0.718312
Source of income or funding	0.717764
Employment status or occupation type	0.715522
Lived within the household for the whole 2019	0.714221
Type of school attended (if applicable)	0.713702
Continuous residence in Italy	0.713145
Secondary Citizenship (if applicable)	0.712773
Italian citizen from birth	0.711452
Quintile in the European Union economic ranking	0.708764

Table C3: Average Importance for variables in KPCA

of the two-cluster assumption by comparing it with alternative models. Specifically, Gaussian Mixture Models with a number of components ranging from one to two were fitted to the score distributions, and model selection criteria (e.g., [Akaike Information Criterion \(AIC\)](#)) were used to assess the optimal number of latent groups. The results consistently favored a two-component solution over a single-component model, supporting the presence of a meaningful separation between typical and atypical observations (see [Table B2](#)). Overall, these analyses indicate that the proposed k -means-based thresholding approach is both stable and robust, providing a reliable and data-driven mechanism for suggesting inlier and outlier observations across different scoring methods.

Appendix C Supporting material for Section 4

Tables pertaining to the 2019 EU-SILC Liguria data analysis and referenced in the main text but omitted for brevity are reported in Appendix.

Selected EU-SILC Variables	Average Importance (AE)
Employment status or occupation type	0.204641
Source of income or funding	0.176416
Respondent's age	0.115671
Number of individuals in the household	0.110805
Indicator of low work intensity	0.080468
Type of school attended (if applicable)	0.076208
Civil status	0.054061
Number of minor Children within the household	0.049749
Type of family	0.042771
Number of women aged 15-55	0.016658
Lived within the household for the whole 2019	0.005160
Quintile in the European Union economic ranking	0.004485
Italian Citizenship	0.001281
Secondary Citizenship (if applicable)	0.000814
Italian citizen from birth	0.000762
Risk of poverty indicator	0.000190
Severity of material deprivation	0.000035
Continuous residence in Italy	0.000035
Social exclusion due to poverty	0.000000

Table C4: Average Importance for variables in AE

Selected EU-SILC Variables	Subgroup 1	Subgroup 2	Subgroup 3	Subgroup 4
Respondent's age	0-16	Over 64	16-32	32-64
Italian Citizenship	YES	YES	YES	YES
Italian citizen from birth	YES	YES	NO	YES
Secondary Citizenship (if applicable)	NO	NO	NO	NO
Continuous residence in Italy	Not applicable	Not applicable	Not applicable	Not applicable
Type of school attended (if applicable)	Upper secondary school	Not applicable	Not applicable	Not applicable
Lived within the household for the whole 2019	YES	YES	YES	YES
Employment status or occupation type	Not applicable	Retired	Other	Employed
Source of income or funding	Not applicable	Pensions	Support from cohabiting family members	Income from salaried employment
Number of individuals in the household	3	2	3	3
Number of minor children within the household	1	0	0	0

Number of women aged 15–55	1	0	1	1
Civil status	Single	Widowed	Single	Single
Type of family	Couple with at least one minor child	Single aged 65 and over	Single parent with only adult children	Couple with only adult children
Severity of material deprivation	Not applicable	Not applicable	Not applicable	Not applicable
Risk of poverty indicator	Not applicable	Not applicable	YES	Not applicable
Indicator of low work intensity	NO	Not applicable	NO	NO
Social exclusion due to poverty	NO	NO	YES	NO
Quintile in the European Union economic ranking	3rd	5th	1st	3rd

Table C5: Medoids for each subgroup based on the analysis through [KPCA](#) of EU-SILC variables

Appendix D Simulation Study

This appendix presents the simulation study presented in [Section 5](#).

D.1 Population and Domain Construction

A synthetic population of $N = 3000$ units was generated from a bivariate Gaussian distribution with mean vector $\mu = [\mu_y, \mu_x]$ and covariance matrix

$$\Sigma = \begin{bmatrix} \sigma_y^2 & \rho\sigma_y\sigma_x \\ \rho\sigma_y\sigma_x & \sigma_x^2 \end{bmatrix},$$

truncated to positive values. Parameter values were set to $\mu_x = 4$, $\mu_y = 1$, $\sigma_x = 0.3$, $\sigma_y = 0.2$, and $\rho = 0.85$. Note that the choice to restrict this simulation to solely positive values ensures that all simulated variables are consistent with the nature of the quantities typically encountered in the application considered. A visual depiction of the synthetic population is provided in [Figure D2](#).

Selected Variables	EU-SILC	Subgroup 1	Subgroup 2	Subgroup 3	Subgroup 4	Subgroup 5	Subgroup 6	Subgroup 7
Respondent's age		16-32	16-32	Over 64	32-64	32-64	32-64	32-64
Italian Citizenship		YES	YES	YES	YES	YES	NO	YES
Italian citizen from birth		YES	YES	YES	YES	NO	Not applicable	YES
Secondary Citizenship (if applicable)		NO	NO	NO	NO	YES	NO	NO
Continuous residence in Italy		Not applicable	Not applicable	Not applicable	Not applicable	YES	YES	Not applicable
Type of school attended (if applicable)		Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Lived within the household for the whole 2019		YES	YES	YES	YES	YES	YES	YES
Employment status or occupation type		Employed	Employed	Retired	Employed	Other	Employed	Employed
Source of income or funding		Support from cohabiting family members	Support from cohabiting family members	Pensions	Income from salaried employment	Support from cohabiting family members	Support from cohabiting family members	Support from cohabiting family members
Number of individuals in the household		4	4	6	3	9	3	2
Number of minor Children within the household		0	0	2	0	3	1	0
Number of women aged 15-55		1	1	1	1	4	1	0
Civil status		Single	Single	Married (living with spouse)	Divorced	Married (living with spouse)	Married (living with spouse)	Married (living with spouse)
Type of family		Couple with only adult children	Couple with only adult children	Two or more family units	Single aged 35-64	Two or more family units	Couple with at least one minor child	Couple without children – Woman aged 35-64
Severity of material deprivation		Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Risk of poverty indicator		YES	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Indicator of low work intensity		NO	NO	Not Applicable	NO	NO	NO	Not Applicable
Social exclusion due to poverty		YES	NO	NO	NO	NO	NO	NO
Quintile in the European Union economic ranking		1st	5th	2nd	4th	2nd	4th	5th

Table C6: Medoids for each subgroup based on the AE analysis of EU-SILC variables

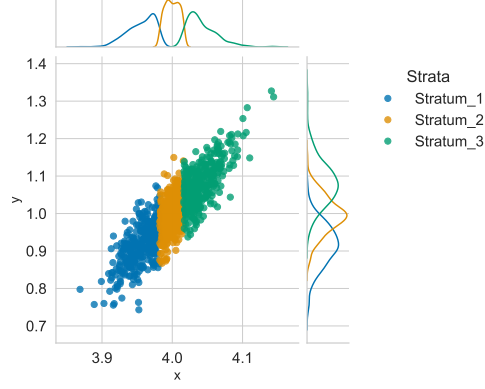


Fig. D2: Scatter plot of the synthetic bivariate Gaussian population, stratified by sampling frame. Marginal distributions of each variable are shown along the axes for each stratum.

The y -variable was treated as the survey response. For the multi-frame setting, units were sequentially allocated into $D = 2^Q - 1$ disjoint domains, according to pre-specified domain sizes. Domains were subsequently aggregated into $Q = 3$ frames based on inclusion indicators, ensuring that the union of frames reproduced the full population while allowing for controlled frame overlap. In contrast, for the [STS](#) setting, we also considered $Q = 3$ strata, in order to facilitate interpretability and comparability of results across the two designs.

D.2 Sampling and Allocation

The two sampling strategies differed in both design and implementation. For the [STS](#) case, the most straightforward setting, simple random sampling without replacement, was employed within each stratum. By contrast, in the multi-frame setting, Sampford sampling was used to select units within each frame, ensuring valid inclusion probabilities irrespective of estimator choice. Two allocation schemes were examined. In the proportional allocation scheme, sample sizes were assigned in proportion to frame sizes,

$$n_q = n \cdot \frac{N_q}{\sum_{q=1}^Q N_q}, \quad \text{for } q = 1, \dots, Q$$

where n is the overall sample size and N_q the population size of frame q . This allocation is computationally simple and effective when the variability across frames is homogeneous.

The second scheme was a cost-optimal allocation under a fixed total survey budget C , with a linear cost function

$$C = c_0 + \sum_{q=1}^Q n_q c_q,$$

where c_0 denotes fixed costs and c_q the per-unit cost of surveying frame q . The optimal frame-specific sample sizes are

$$n_q^{opt} = \frac{(C - c_0)N_q\sigma_{q\alpha}/\sqrt{c_q}}{\sum_{q=1}^Q N_q\sigma_{q\alpha}\sqrt{c_q}},$$

where $\sigma_{q\alpha}$ is the frame-specific standard deviation adjusted for unit multiplicity. This allocation maximises estimator precision given the cost constraint [19, 28]. In practice, $\sigma_{q\alpha}$ may be estimated from historical data, expert judgment, or pilot studies. For convenience, we assumed no fixed cost ($c_0 = 0$) and considered frame-specific per-unit costs that were approximately equal but with slight differences. Specifically, we set $c_1 = 9.5$, $c_2 = 10$, and $c_3 = 10.5$. For the stratified case, the multiplicity adjustment is not needed, leading to the simplified expression [21],

$$n_q^{opt} = \frac{(C - c_0)N_q\sigma_q/\sqrt{c_q}}{\sum_{q=1}^Q N_q\sigma_q\sqrt{c_q}},$$

where σ_q is the within-stratum standard deviation. In realistic applications, frame or stratum variances are rarely known a priori and must be estimated. In the simulation setting, however, the true values were employed, since the entire synthetic population was available.

D.3 Estimation Methods

For the multi-frame design, two estimators were evaluated. The first one was the Simple Multiplicity (SM) estimator, which adjusts for unit multiplicity m_{kq} ,

$$\hat{Y}_{SM} = \sum_{q=1}^Q \sum_{k \in s_q} y_k m_{kq}^{-1} \pi_{kq}^{-1},$$

where π_{kq} is the first-order inclusion probability, and units in the same domain share the same multiplicity; m_{kq} denotes the multiplicity of the k -th individual within the q -th frame; see Mecatti and Singh [24].

The second estimator was the Pseudo Maximum Likelihood (PML) estimator, which accounts for frame overlap and domain allocation by iteratively adjusting domain-frame totals. This method, originally proposed by Lohr and Rao [23], provides an optimal linear estimator of the population total when samples are drawn independently from each frame using probability sampling designs.

For the stratified design, the classical stratified estimator was applied, which is defined as the weighted sum of within-stratum means, with weights equal to the population stratum proportions.

D.4 Monte Carlo Evaluation

Each simulation scenario was replicated N_{run} times, using a fixed random seed to ensure reproducibility. Performance was assessed through the MSE (see Section 5.2).

For each estimator and allocation scheme, mean estimates and 95% bootstrap confidence intervals were reported. Additional evaluation criteria, such as variance reduction and cost efficiency, are discussed in the main text.

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